

Lecture 14

MLE, MAP and Bias-Variance Trade-off

EECS 189/289, Spring 2026 @ UC Berkeley
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Roadmap

- MLE Recap: Least Squares as Maximum Likelihood
- Prior Beliefs
- MLE vs MAP
- Bias-Variance Trade-off



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MLE Recap: Least Squares as Maximum Likelihood

- MLE Recap: Least Squares as Maximum Likelihood
- Choosing Different Noise Models
- Prior Beliefs
- Bias-Variance Trade-off



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MLE vs MAP. Estimating bias of a coin

- Say Y is the unknown success probability of a coin. Y in $[0,1]$.
- x_1, x_2 observed binary outcomes.
- $P(x_1=1 | Y) = Y$. alternatively, $P(x_1=0 | Y) = (1-Y)$.



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MLE vs MAP. Estimating bias of a coin

- Say Y is the unknown success probability of a coin. Y in $[0,1]$.
- x_1, x_2 observed binary outcomes.
- $P(x_1=1 | Y) = Y$. alternatively, $P(x_1=0 | Y) = (1-Y)$.
- $P(x_1 | Y) = Y^{x_1} (1-Y)^{1-x_1}$
- Bayesian idea: Prior for Y :
(let's use a discrete Prior, you can also use a Beta prior)
- $P(Y=0.9) = P(Y=0.1) = 0.5$



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MLE vs MAP

- Say observed dataset $D = (x_1, x_2) = (1, 1)$.
 - Find MLE and MAP estimates for Y .
 - Likelihood $P(D | Y) = \prod_{i=1}^2 Y^{x_i} (1 - Y)^{1-x_i}$
 - So Likelihood of $D = (1, 1)$: $P(D | Y) = Y^2$
 - Derivative of Log-likelihood: $\frac{d}{dY} \log P(D|Y) = \frac{d}{dY} \log(Y^2) = \frac{2}{Y}$.
 - So $\hat{Y}_{MLE} = 1$.
 - Find $\hat{Y}_{Map}$ and compare.
-
- Posterior $P(Y, x_1, x_2) = P(Y)P(x_1 | Y)P(x_2 | Y)$

MLE vs MAP



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- $\hat{Y}_{Map} = \operatorname{argmax} P(Y = y|D).$
- $P(Y | D) = \frac{P(D|Y)P(Y)}{P(D)}.$
- $P(D) = P(D | Y=0.1)P(Y=0.1) + P(D | Y=0.9)P(Y=0.9)$
- $P(D | Y=0.1) = 0.1^2 = 0.01. \quad P(D | Y=0.9) = 0.9^2 = 0.81.$
- $P(D) = 0.01 * 1/2 + 0.81 * 1/2 = 0.41.$
- So $P(Y=0.9 | D) P(Y=0.9) = 0.81 * 0.5$
- $P(Y=0.1 | D) P(Y=0.1) = 0.1 * 0.5$
So MAP estimate is $\hat{Y}_{Map} = 0.9.$



Belief about Parameters

- Just like coins: we can encode a prior belief about weights $\mathbf{w}$.
- Common prior: $w_j \sim \mathcal{N}(0, \sigma_w^2)$
 - Centered at 0 $\rightarrow$ no strong effect unless data supports it.
 - Variance σ_w^2 controls how strongly we expect small weights.
- This prior = belief that simpler models are preferred.

Likelihood function

$$p(\mathbf{t} | \mathbb{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)$$

Prior (small, centered)

$$p(\mathbf{w} | \tau^2) = \frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|\mathbf{w}\|^2}{2\tau^2}}$$

Posterior (from Bayes rule) up to a constant: (We prove this):

$$p(\mathbf{w} | \mathbf{t}) \propto \exp \left[-\frac{1}{2\sigma^2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 - \frac{1}{2\tau^2} \|\mathbf{w}\|^2 \right].$$

What Is MAP?



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$$p(\mathbf{w} | \mathbf{t}) = \frac{p(\mathbf{t} | \mathbf{w}) p(\mathbf{w})}{p(\mathbf{t})}$$

Bayes Rule

$$p(\mathbf{w} | \mathbf{t}) \propto \underbrace{\prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)}_{\text{Likelihood } p(\mathbf{t} | \mathbf{w})} \underbrace{\frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|\mathbf{w}\|^2}{2\tau^2}}}_{\text{Prior } p(\mathbf{w})}$$

Maximum A Posteriori (MAP):

$$\hat{\mathbf{w}}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} | \mathbf{t}) = \underset{\mathbf{w}}{\operatorname{argmax}} [p(\mathbf{t} | \mathbf{w}) \times p(\mathbf{w})]$$



What Is MAP?

Maximum A Posteriori (MAP):

$$\hat{\mathbf{w}}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmax}} p(\mathbf{w} \mid \mathbf{t}) = \underset{\mathbf{w}}{\operatorname{argmax}} [p(\mathbf{t} \mid \mathbf{w}) \times p(\mathbf{w})]$$

Taking $-\log$ of the objective turns maximization problem to a minimization problem:

$$\hat{\mathbf{w}}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} [-\ln p(\mathbf{t} \mid \mathbf{w}) - \ln p(\mathbf{w})]$$



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Plugging in Distributions

$$\begin{aligned}\hat{\mathbf{w}}_{MAP} &= \underset{\mathbf{w}}{\operatorname{argmin}} \left[\underbrace{-\ln p(\mathbf{t} | \mathbf{w})}_{-\ln \prod_{n=1}^N \mathcal{N}(t_n | \mathbf{w}^T \phi(\mathbf{x}_n), \sigma^2)} - \underbrace{\ln p(\mathbf{w})}_{-\ln \frac{1}{(2\pi\tau^2)^{\frac{M}{2}}} e^{-\frac{\|\mathbf{w}\|^2}{2\tau^2}}} \right] \\ &= \boxed{\frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(\mathbf{x}), \mathbf{w}))^2} = \boxed{\frac{\|\mathbf{w}\|^2}{2\tau^2}} + \text{const}\end{aligned}$$

$$\hat{\mathbf{w}}_{MAP} = \underset{\mathbf{w}}{\operatorname{argmin}} \left[\boxed{\frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(\mathbf{x}), \mathbf{w}))^2} + \boxed{\frac{\|\mathbf{w}\|^2}{2\tau^2}} \right]$$

Φ is an $N \times M$ matrix.
(Design Matrix)
Each row is a feature

$$\operatorname{argmin} \left[\frac{1}{2\sigma^2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{1}{2\tau^2} \|\mathbf{w}\|^2 \right].$$

Plugging in Distributions



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$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[\frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(x), w))^2 + \frac{\|w\|^2}{2\tau^2} \right]$$

$$\operatorname{argmin} \left[\frac{1}{2\sigma^2} \|\mathbf{t} - \Phi \mathbf{w}\|^2 + \frac{1}{2\tau^2} \|\mathbf{w}\|^2 \right].$$

$$\begin{array}{c} \text{M features} \\ \begin{array}{|c|} \hline x_1 \\ \hline x_2 \\ \hline \Phi \\ \hline x_N \\ \hline \end{array} \end{array} \quad \begin{array}{c} \text{N samples} \\ \mathbf{t} \end{array} - \begin{array}{c} \mathbf{w} \end{array} \quad \begin{array}{c} \|\mathbf{t} - \Phi \mathbf{w}\|^2 \end{array} + \frac{1}{2\tau^2} \begin{array}{c} \|\mathbf{w}\|^2 \end{array}$$

Φ is an $N \times M$ matrix.
(Design Matrix)
Each row is a feature



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Does This Look Like a Ridge Regression?

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[\frac{1}{2\sigma^2} \sum_n (t_n - y(\phi(x), w))^2 + \frac{\|w\|^2}{2\tau^2} \right]$$

Multiply by σ^2

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[\underbrace{\frac{1}{2} \sum_n (t_n - y(\phi(x), w))^2}_{\frac{1}{2} E_D(w)} + \underbrace{\frac{\sigma^2 \|w\|^2}{2\tau^2}}_{E_w(w)} \right]$$

$w \sim N(0, \tau^2 I)$

λ

Least-Squares + Gaussian Prior $\Rightarrow$ Ridge Regression

MLE, MAP and Bias-Variance Trade-off

- MLE
In Linear regression with Gaussian noise, Least Square minimization=MLE
- If you put a zero-mean gaussian prior on w .
Ridge = MAP.
- Bias-Variance Trade-off



Fundamental Challenges in Learning?



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- ***Fit the Data***
 - Provide an explanation for what we observe
- ***Generalize to the World***
 - Predict the future
 - Explain the unobserved
 - Believe your prior



Is this cat grumpy or are we
overfitting to human faces?



Fundamental Challenges in Learning?

- **Bias:** The expected deviation between the predicted value and the true value.
- **Variance:** Two sources
 - **Noise:** The variability of the random noise in the process we are trying to model.
 - **Model Variance:** The variability in the predicted value across different training datasets.

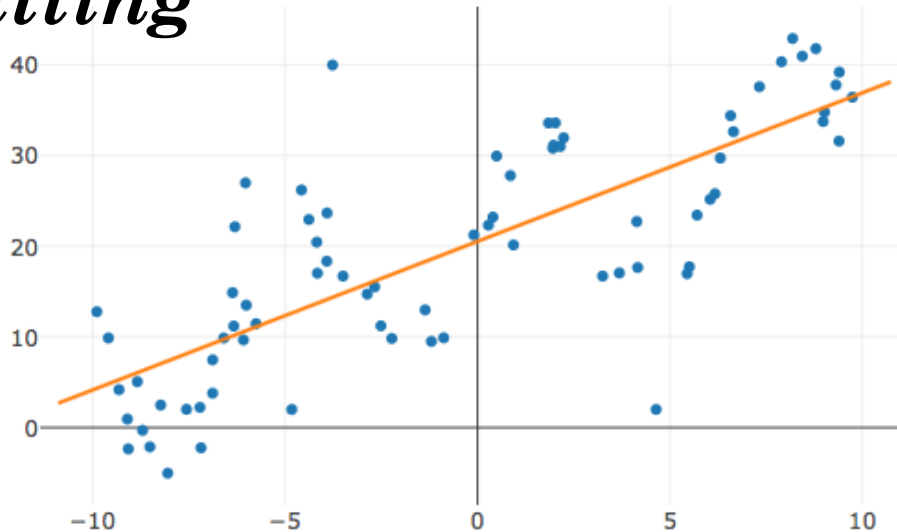
Bias



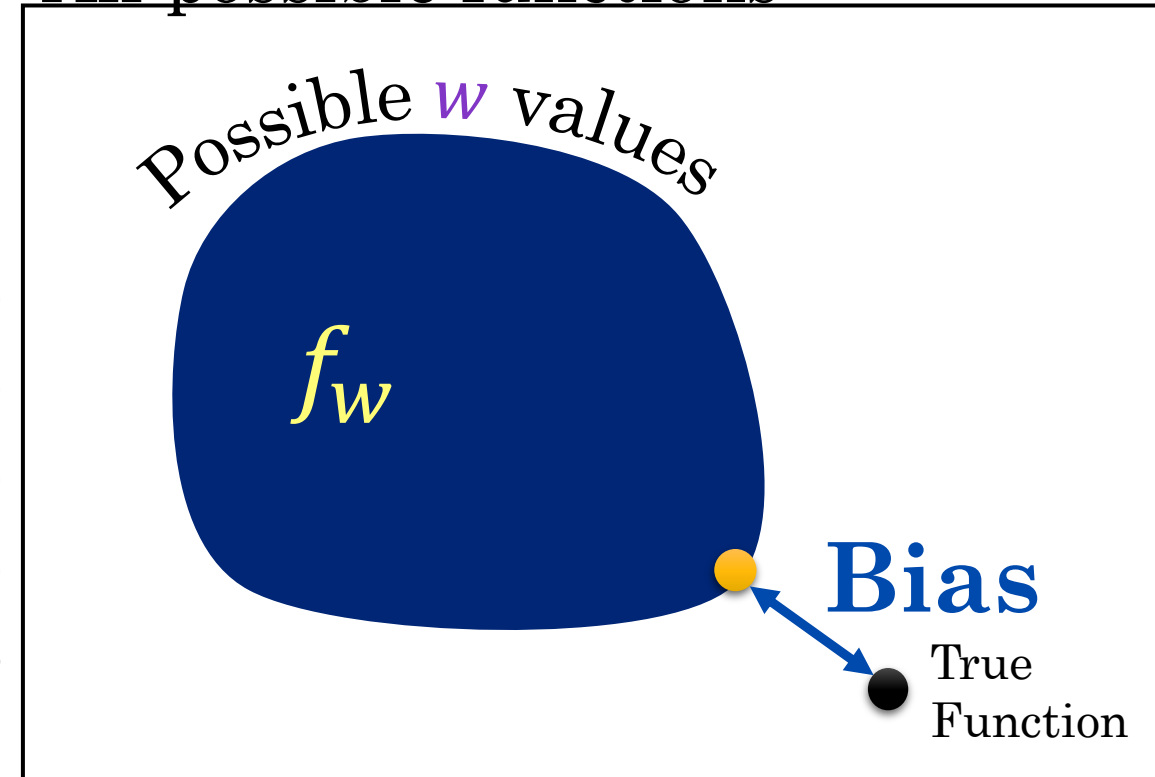
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The expected deviation between the predicted value and the true value

- Depends on both the:
 - Choice of f
 - Learning procedure
- ***Under-fitting***



All possible functions



Noise

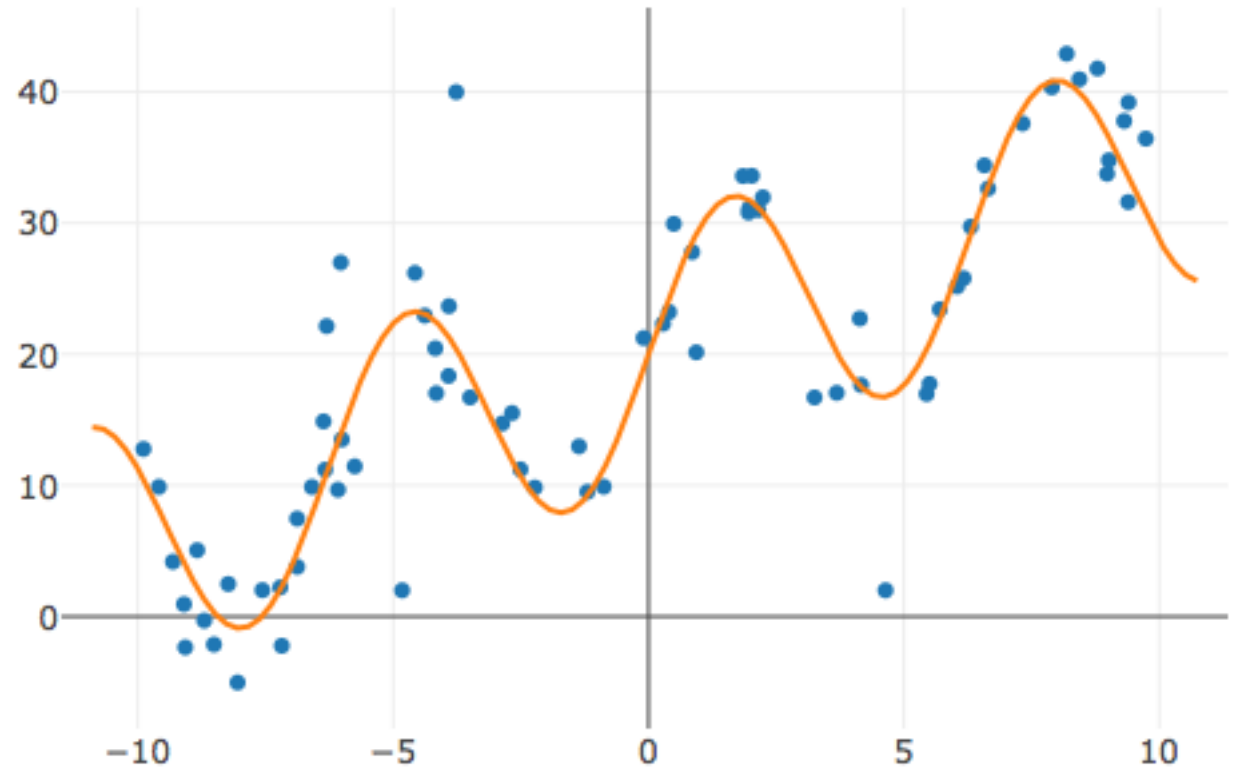


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The variability of the random noise in the process we are trying to model.

- Measurement variability
- Stochasticity
- Missing information

**Beyond our control
(usually)**



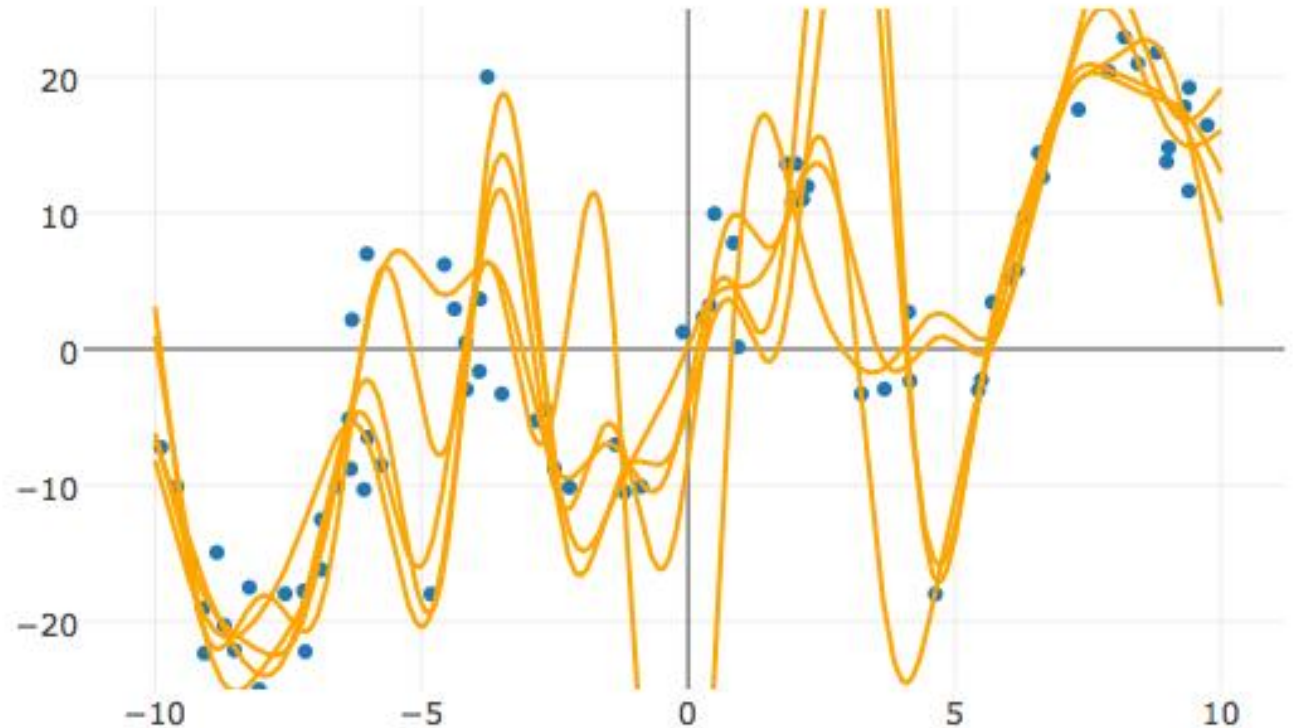


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Model Variance

Variability in the predicted value across different training datasets.

- Sensitivity to variation in the training data
- Poor generalization
- **Overfitting**



Analysis of Squared Error



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- For the test point $\mathbf{x}$ the expected error:
 - *Random variables are red*

Assume noisy observations
→ t is a random variable

True Function

$$t = h(\mathbf{x}) + \epsilon$$

Noise term:

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

Can be any parametric
function
Linear Regression is $\mathbf{w}^T \mathbf{x}$

Assume *training data* is
random
→ w is a random variable

Analysis of Squared Error



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Goal: Expected test error
over repeated datasets.

$$E[(t - f_w(x))^2] =$$

“Noise” + (Bias)² + Model Variance



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$$E[(t - f_w(\mathbf{x}))^2] = E[(t - \underbrace{h(\mathbf{x}) + h(\mathbf{x})}_{\text{Subtracting and adding } h(\mathbf{x})} - f_w(\mathbf{x}))^2]$$

Subtracting and adding $h(\mathbf{x})$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2] = E[\underbrace{(t - h(\mathbf{x}))}_a + \underbrace{h(\mathbf{x}) - f_w(\mathbf{x}))}_b]^2]$$

Expanding in terms of a and b : $(a + b)^2 = a^2 + b^2 + 2ab$

$$\begin{aligned} &= E[(t - h(\mathbf{x}))^2] + E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2] \\ &\quad + 2 E[\underbrace{(t - h(\mathbf{x}))}_\epsilon \underbrace{(h(\mathbf{x}) - f_w(\mathbf{x}))}_{2ab}] \\ &\quad + 2 E[\epsilon (h(\mathbf{x}) - f_w(\mathbf{x}))] \end{aligned}$$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2] = E[\underbrace{(t - h(\mathbf{x}))}_a + \underbrace{h(\mathbf{x}) - f_w(\mathbf{x})}_b]^2]$$

Expanding in terms of a and b : $(a + b)^2 = a^2 + b^2 + 2ab$

$$= E[(t - h(\mathbf{x}))^2] + E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2] + 2 E[\epsilon (h(\mathbf{x}) - f_w(\mathbf{x}))]$$

Independence of ϵ and w

$$+ 2 \underbrace{E[\epsilon]}_0 E[(h(\mathbf{x}) - f_w(\mathbf{x}))]$$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2]$$

$$= \sigma^2$$

$$= E[(t - h(\mathbf{x}))^2] +$$

Obs. Value

True Value

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

True Value

Pred. Value

“*Noise*” Term

Model
Estimation
Error

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$



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$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \sigma^2$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model
Estimation
Error

We need to calculate
this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \sigma^2$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model
Estimation
Error

We need to calculate
this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$



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$$\mathbb{E} \left[\left(h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] = \text{Next we will show...}$$

$$\underbrace{\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2}_{(\text{Bias})^2} + \underbrace{\mathbb{E} \left[\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right]}_{\text{Model Variance}}$$

➤ How?

➤ Adding and subtracting what?



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$$\mathbb{E} \left[\left(h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] =$$

$$\mathbb{E} \left[\underbrace{\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_a + \underbrace{\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)}_b \right]^2$$

Expanding in terms of a and b : $(a + b)^2 = a^2 + b^2 + 2ab$

$$\begin{aligned} & \mathbb{E} \left[\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right] + \mathbb{E} \left[\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] + \\ & 2\mathbb{E} \left[\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) \left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right] \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[\left(h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] &= \\ \mathbb{E} \left[\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right] &+ \mathbb{E} \left[\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] + \\ 2 \underbrace{\mathbb{E} \left[\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) \left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right]}_{\text{constant}} & \\ \underbrace{2 \left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) \mathbb{E} \left[\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right]}_{\text{constant}} & \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[\left(h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] &= \\ \mathbb{E} \left[\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right] &+ \mathbb{E} \left[\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] + \\ 2 \left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right) &\underbrace{\mathbb{E} \left[\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right) \right]}_{\text{constant}} \\ \underbrace{2 \left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_{\text{constant}} &\underbrace{\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)}_0 \end{aligned}$$



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$$\begin{aligned} \mathbb{E} \left[\left(h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] &= \\ \underbrace{\mathbb{E} \left[\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2 \right]}_{\text{constant}} &+ \mathbb{E} \left[\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] \\ (h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})])^2 &+ \end{aligned}$$



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$$\mathbb{E} \left[\left(h(\mathbf{x}) - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right] =$$
$$\underbrace{\left(h(\mathbf{x}) - \mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] \right)^2}_{(\text{Bias})^2} + \underbrace{\mathbb{E} \left[\left(\mathbb{E}[f_{\mathbf{w}}(\mathbf{x})] - f_{\mathbf{w}}(\mathbf{x}) \right)^2 \right]}_{\text{Model Variance}}$$

$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \sigma^2$$

“*Noise*” Term

$$E[(h(\mathbf{x}) - f_w(\mathbf{x}))^2]$$

Model
Estimation
Error

We now have
calculated this term

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$Var(\epsilon) = \sigma^2$$

$$E[(t - f_w(\mathbf{x}))^2]$$

$$= E[(t - h(\mathbf{x}))^2] + \sigma^2$$

“*Noise*” Term

$$(h(\mathbf{x}) - E[f_w(\mathbf{x})])^2 + (\text{Bias})^2$$

$$E[(E[f_w(\mathbf{x})] - f_w(\mathbf{x}))^2] \quad \text{Model Variance}$$

Useful Equations:

$$t = h(\mathbf{x}) + \epsilon$$

$$E(\epsilon) = 0$$

$$\text{Var}(\epsilon) = \sigma^2$$



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$$\begin{aligned} & \mathbb{E}[(t - f_w(\mathbf{x}))^2] \\ &= \mathbb{E}[(t - h(\mathbf{x}))^2] + \quad \text{“Noise” Term} \\ & \quad (h(\mathbf{x}) - \mathbb{E}[f_w(\mathbf{x})])^2 + \quad \text{(Bias)}^2 \\ & \quad \mathbb{E}[(\mathbb{E}[f_w(\mathbf{x})] - f_w(\mathbf{x}))^2] \quad \text{Model} \\ & \quad \text{Variance} \end{aligned}$$



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Does This Look Like a Ridge Regression?

$$\hat{w}_{MAP} = \underset{w}{\operatorname{argmin}} \left[\underbrace{\frac{1}{2} \sum_n (t_n - y(\phi(x), w))^2}_{\frac{1}{2} E_D(w)} + \underbrace{\frac{\sigma^2 \|w\|^2}{2\tau^2}}_{E_w(w)} \right] \quad \begin{array}{l} w \sim N(0, \tau^2 I) \\ \text{prior} \end{array}$$

λ

Increase λ : weights shrink $\rightarrow$ Model less flexible. Bias increases. But shrinking makes $\hat{w}_{MAP}$ less sensitive to noise. So variance decreases. After a point you tend to underfit.

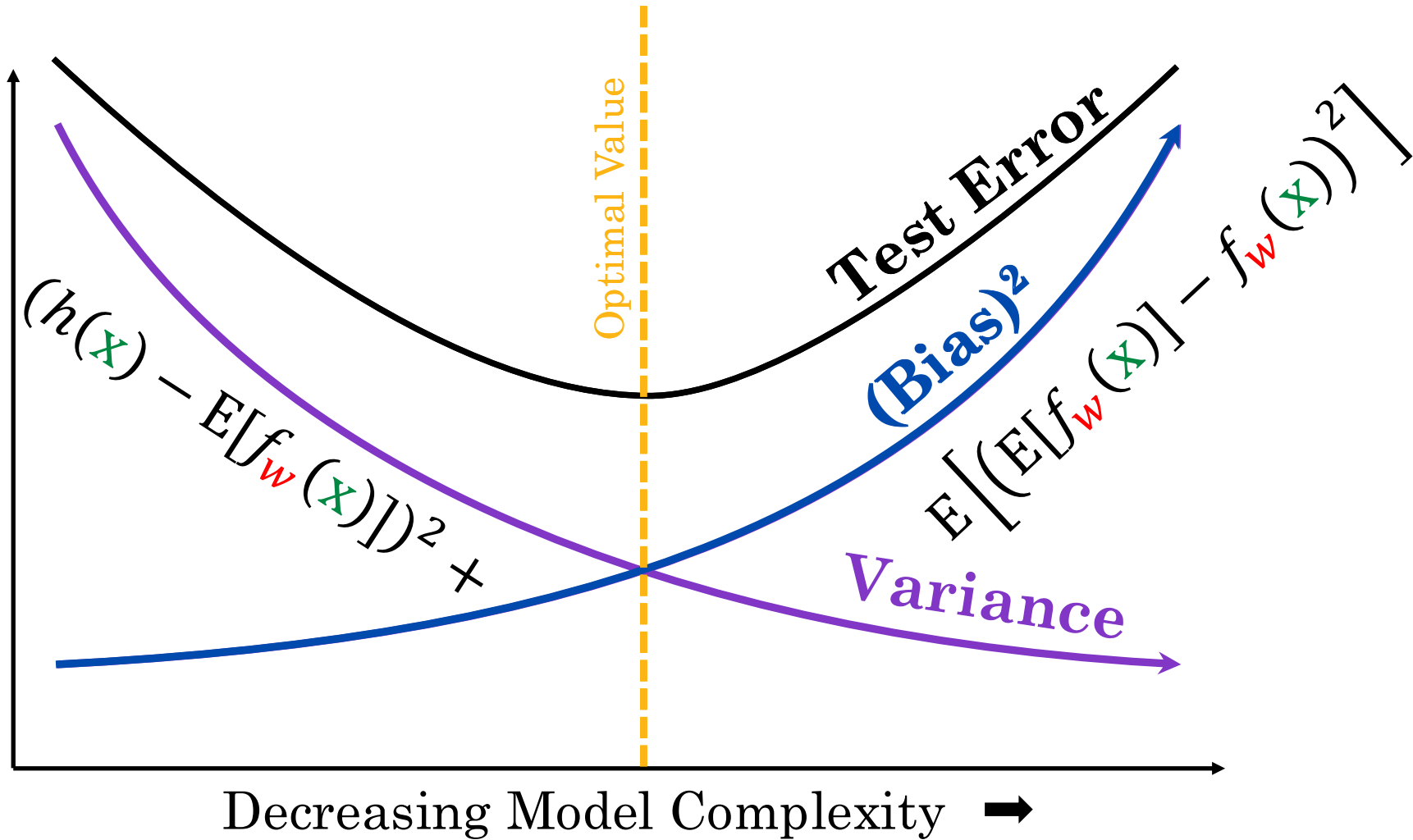
As λ goes to 0: model more flexible. Bias decreases, variance increases. You tend to overfit.

So the regularization term is a mechanism that trades variance for bias.
Overfit = high variance. Underfit = high bias.

Bias Variance Plot



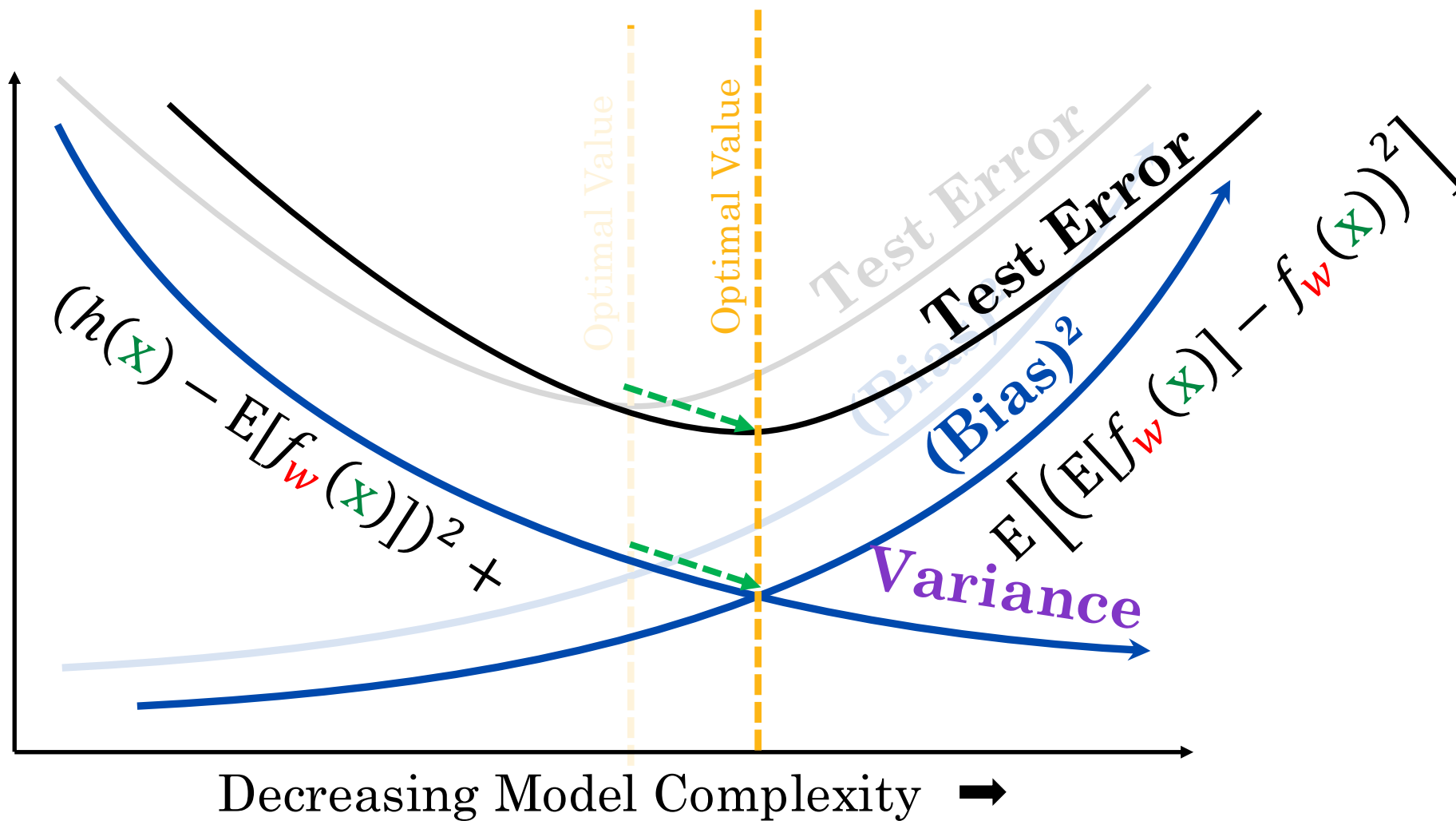
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More Data Supports More Complexity



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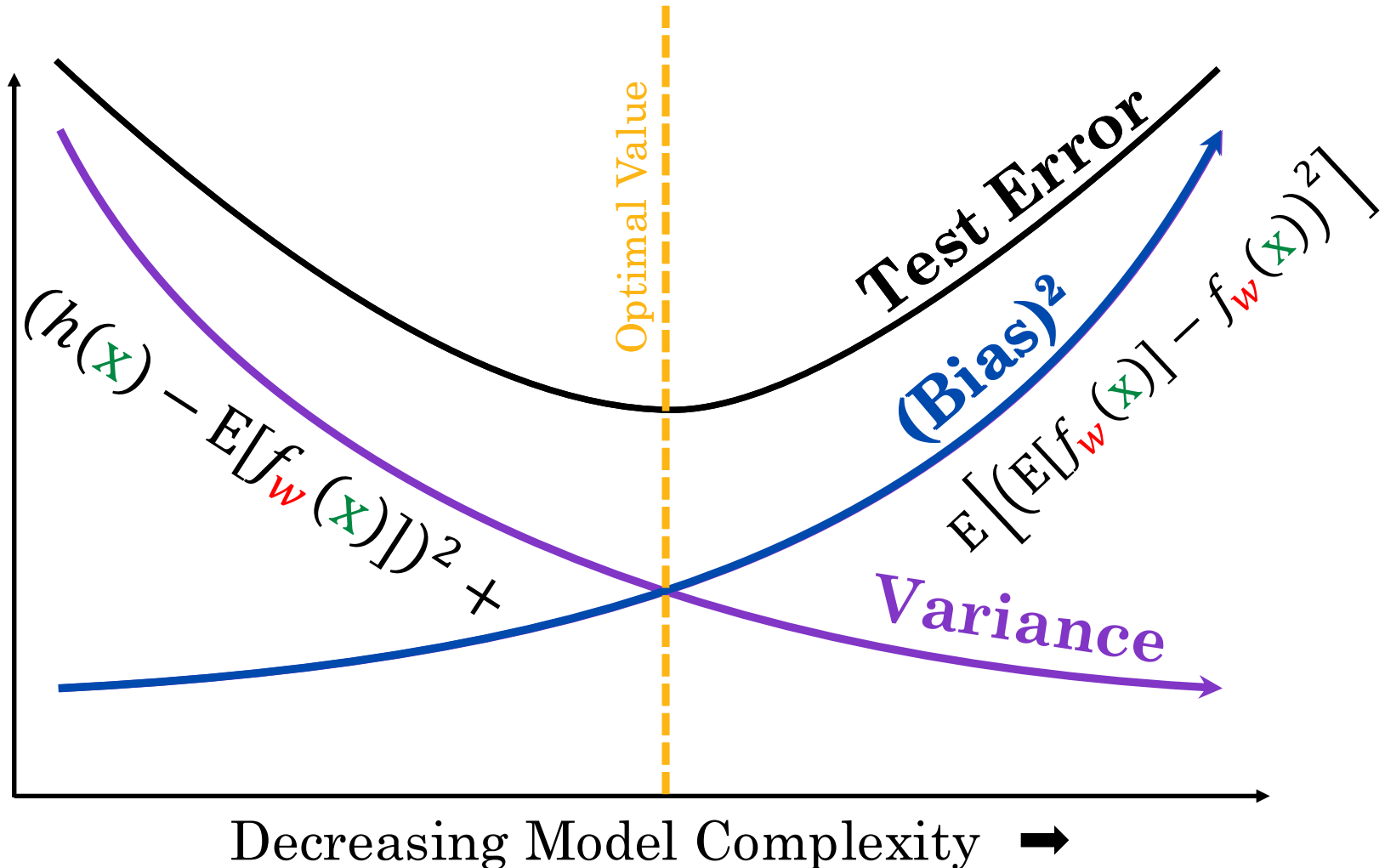


How Do We Control Model Complexity?

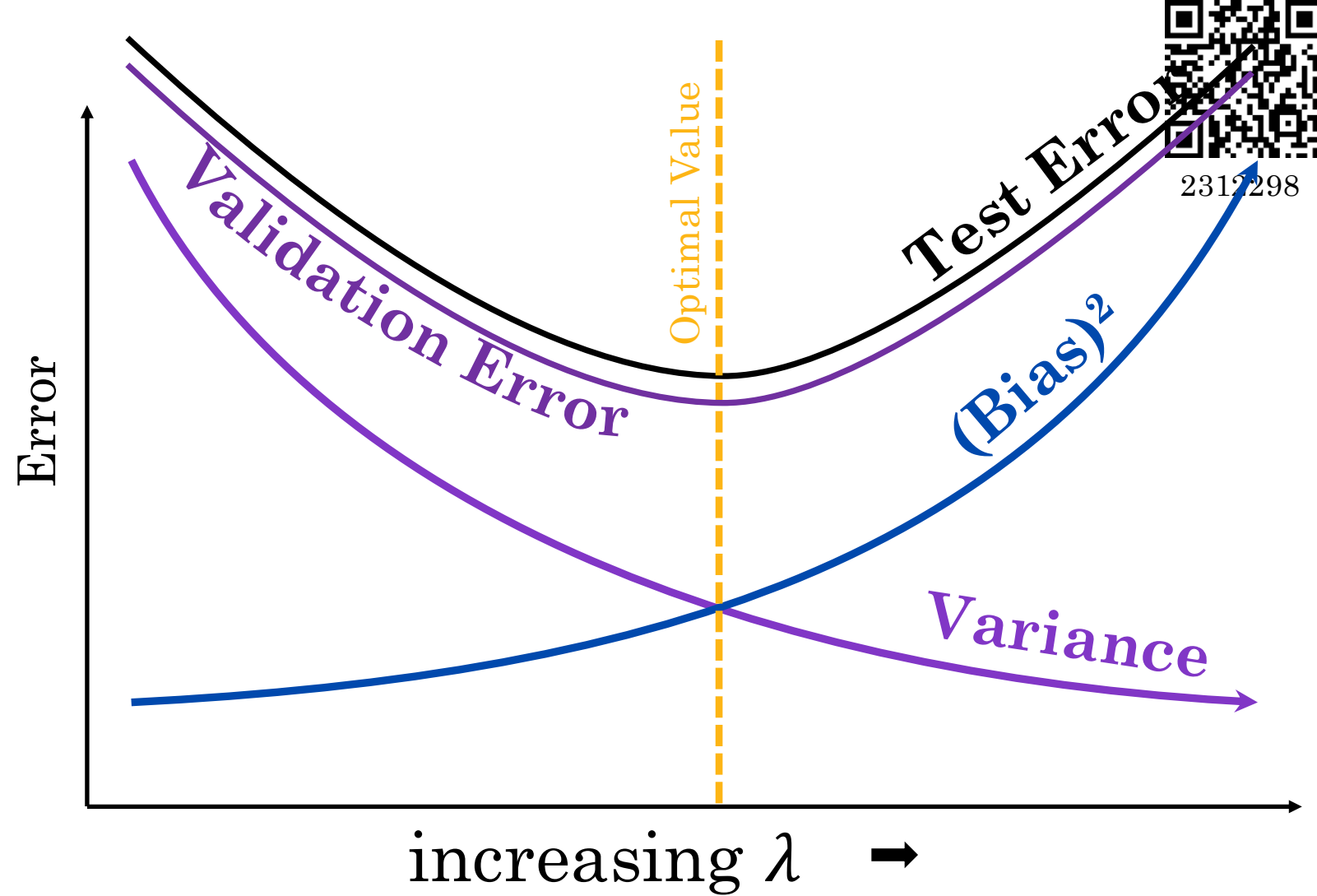


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- So far:
 - Number of features
 - Choices of features
 - **Regularization**



Determining the Optimal λ



- Value of λ determines bias-variance tradeoff
 - Larger values $\rightarrow$ more regularization $\rightarrow$ more bias $\rightarrow$ less variance
- Determined through validation



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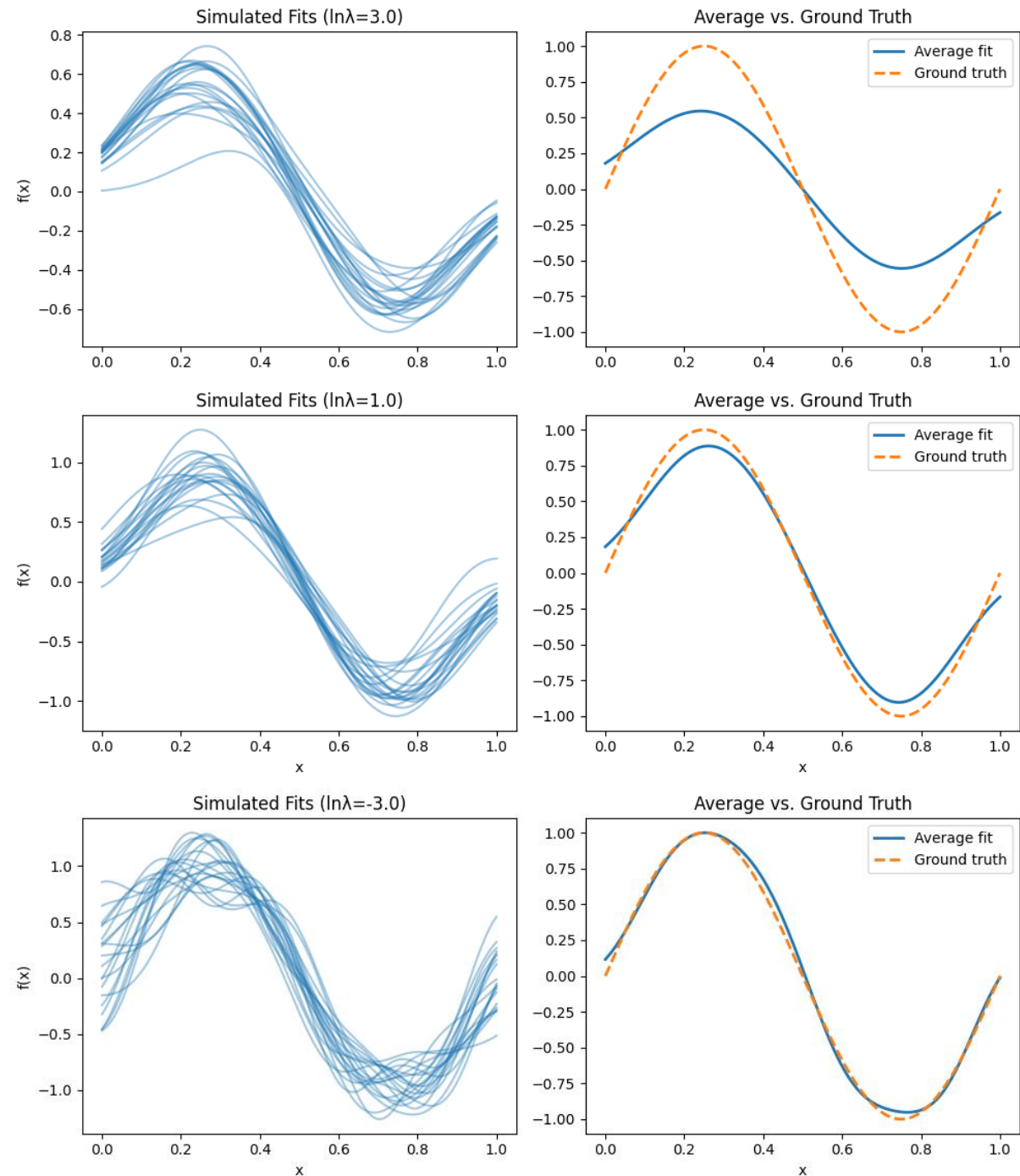
Dataset Example

Data generation function:
 $h(x) = \sin(2\pi x)$, $x \sim \text{Uniform}(0,1)$

Generate L independent datasets ($L = 100$), each with $N = 25$ noisy observations.

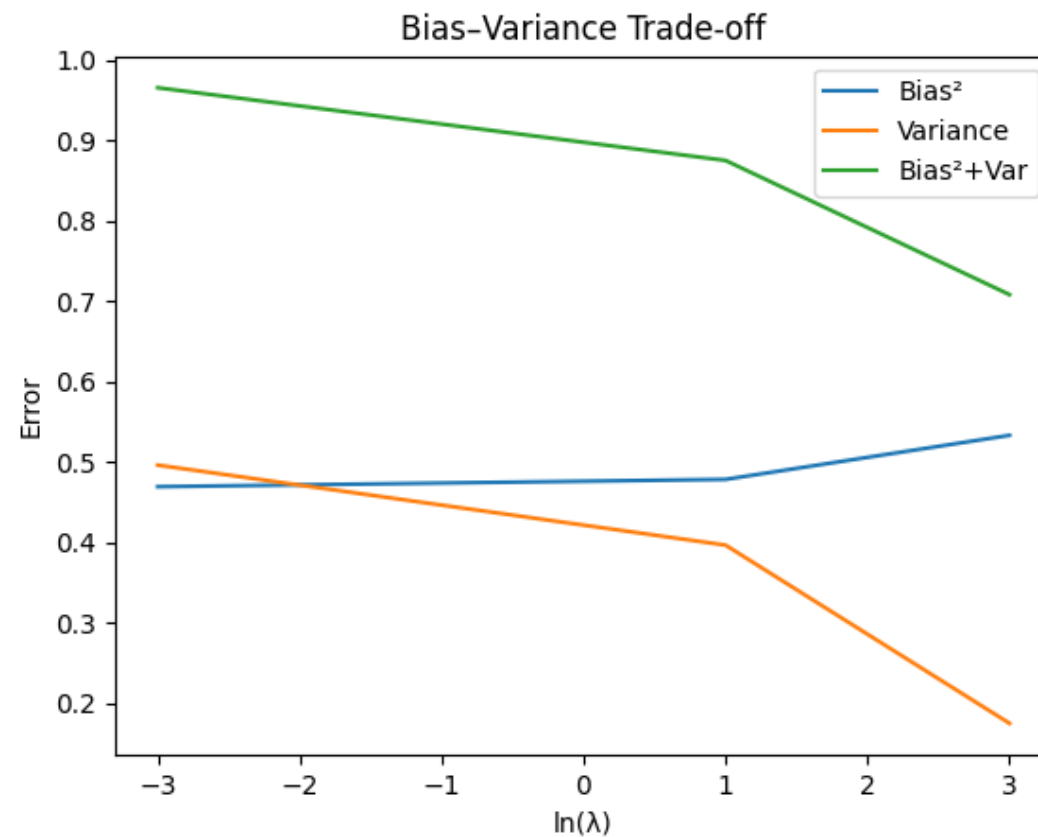
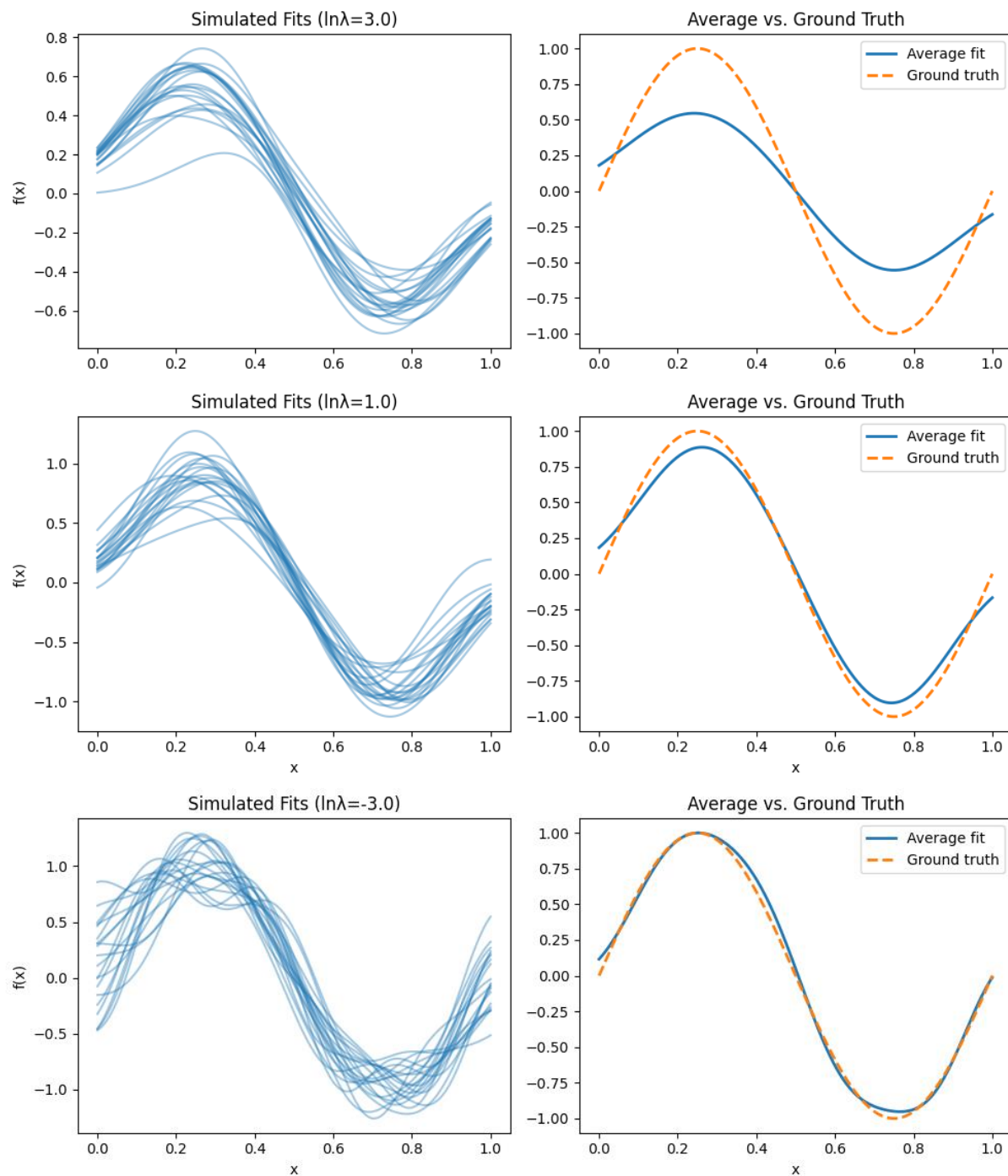
$$t_n = h(x_n) + \epsilon_n,$$
$$\epsilon_n \sim \mathcal{N}(0, 0.32)$$

Model: linear combination of $M = 24$ Gaussian basis functions plus bias.





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Does T

$$\hat{w}_{MAP} =$$

As $\lambda \rightarrow 0$ (less regularization):

Train error: ?

Test error: ?

As λ increases (stronger regularization):

Train error: ?

Test error: ?

h?



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$w \sim N(0, \tau^2 I)$
prior

λ

Increase λ : weights shrink $\rightarrow$ Model less flexible. Bias increases. But shrinking makes $\hat{w}_{MAP}$ less sensitive to noise. So variance decreases. After a point you tend to underfit.

As λ goes to 0: model more flexible. Bias decreases, variance increases. You tend to overfit.

So the regularization term is a mechanism that trades variance for bias.

Overfit = high variance. Underfit = high bias.

Does This Look Like a Ridge Regression?



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$$\hat{w}_{MAP} =$$

As $\lambda \rightarrow 0$ (less regularization):

Train error: decreases.

Test error: typically decreases at first, but can increase when you start overfitting (U-shape).

As λ increases (stronger regularization):

Train error: increases (worse fit).

Test error: typically decreases at first (less overfitting), then increases when λ is too large and you underfit (also U-shape).

Increase λ : we
less sensitive to

So: train error is increasing in λ

Test error is usually U-shaped with a sweet spot at an intermediate λ

As λ goes to 0:

$w \sim N(0, \tau^2 I)$
prior

λ

$\hat{w}_{MAP}$

erfit.

So the regularization term is a mechanism that trades variance for bias.
Overfit = high variance. Underfit = high bias.

Bias Variance Derivation Quiz



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What is the equivalent of $E[t]$?

- A. 0
- B. Bias
- C. Variance
- D. Noise
- E. $h(\mathbf{x})$
- F. $h(\mathbf{x}) + \epsilon$

Bias Variance Derivation Quiz



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What is the equivalent of $E[\epsilon^2]$?

- A. 0
- B. Bias
- C. Variance
- D. Noise
- E. $h(\mathbf{x})$
- F. $h(\mathbf{x}) + \epsilon$

Bias Variance Derivation Quiz



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What is the equivalent of $E \left[\left(E[f_{\textcolor{red}{w}}(\textcolor{green}{x})] - f_{\textcolor{red}{w}}(\textcolor{green}{x}) \right)^2 \right]$?

- A. 0
- B. Bias
- C. Variance
- D. Noise
- E. $h(\textcolor{green}{x})$
- F. $h(\textcolor{green}{x}) + \textcolor{red}{\epsilon}$



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Research Side

Evaluating LMM's in the Wild with



LMarena

An open platform for
human preference evals

LMarena (Formerly Chatbot Arena)

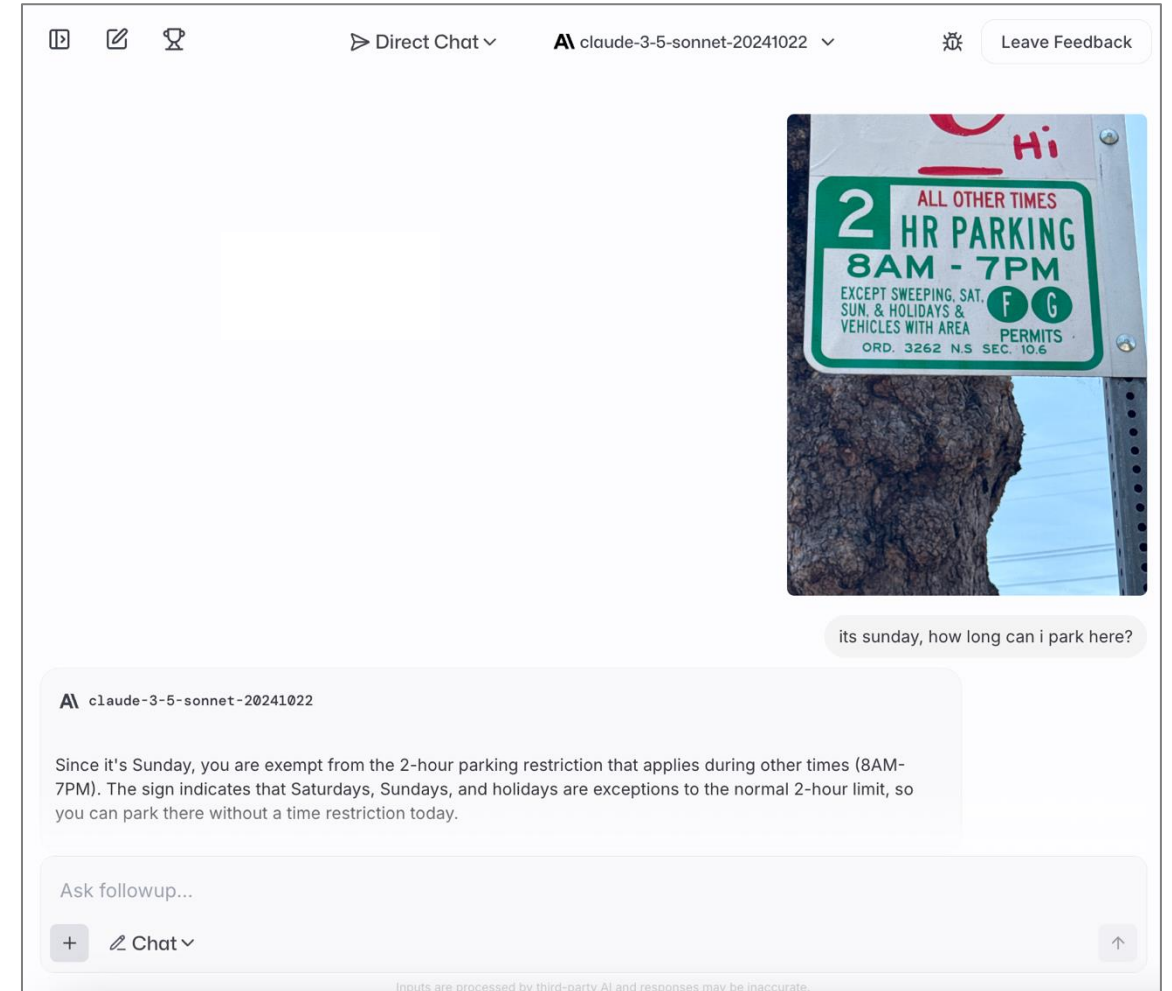
A platform for **holistic LMM evaluation**, where **real user conversations and pairwise votes** are crowdsourced to build a **live human preference leaderboard**.

Members (pre-company launch): Wei-Lin Chiang, Anastasios Angelopoulos, Lianmin Zheng, Ying Sheng, Lisa Dunlap, Chris Chou, Tianle Li, Evan Frick, Aryan Vichare, Naman Jain, Manish Shetty, Yifan Song, Kelly Tang, Sophie Xie, Connor Chen, Joseph Tennyson, Dacheng Li, Siyuan Zhuang, Valerie Chen, Wayne Chi
Advisors (pre-company launch): Ion Stoica, Joseph Gonzalez, Hao Zhang, Trevor Darrell

Crowdsourcing user interactions with LMM's

Direct Chat

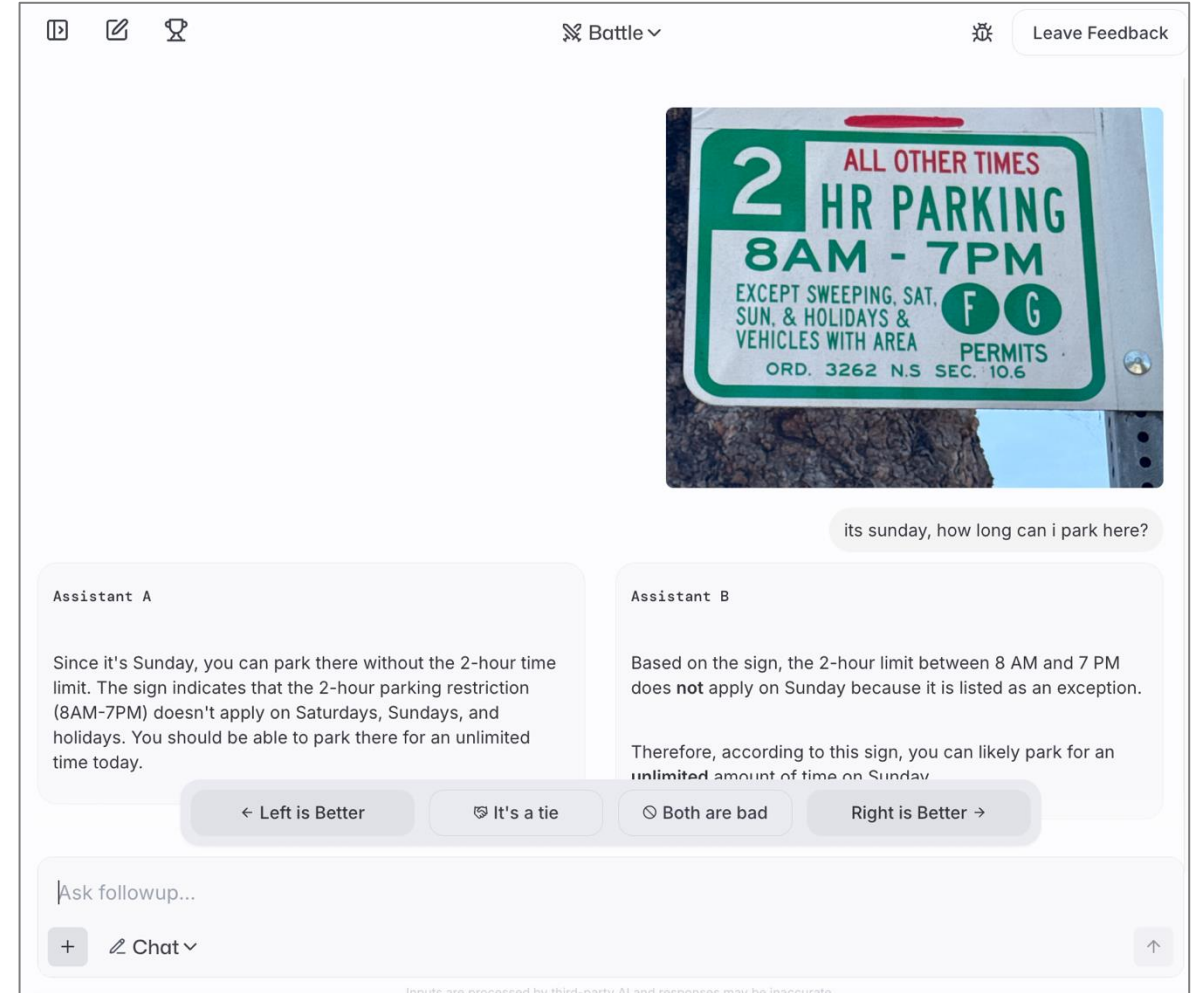
1. User selects a model to **chat** with (LLM, VLM, Text-2-Img, etc)
2. Conversations are recorded to **analyze** real world interactions



Crowdsourcing user interactions with LMM's

Battle

1. User input **any prompt**
 2. **Two** anonymized models give answers side-by-side
 3. User **vote** which is best
- Pairwise setting allows for more fine grained comparison





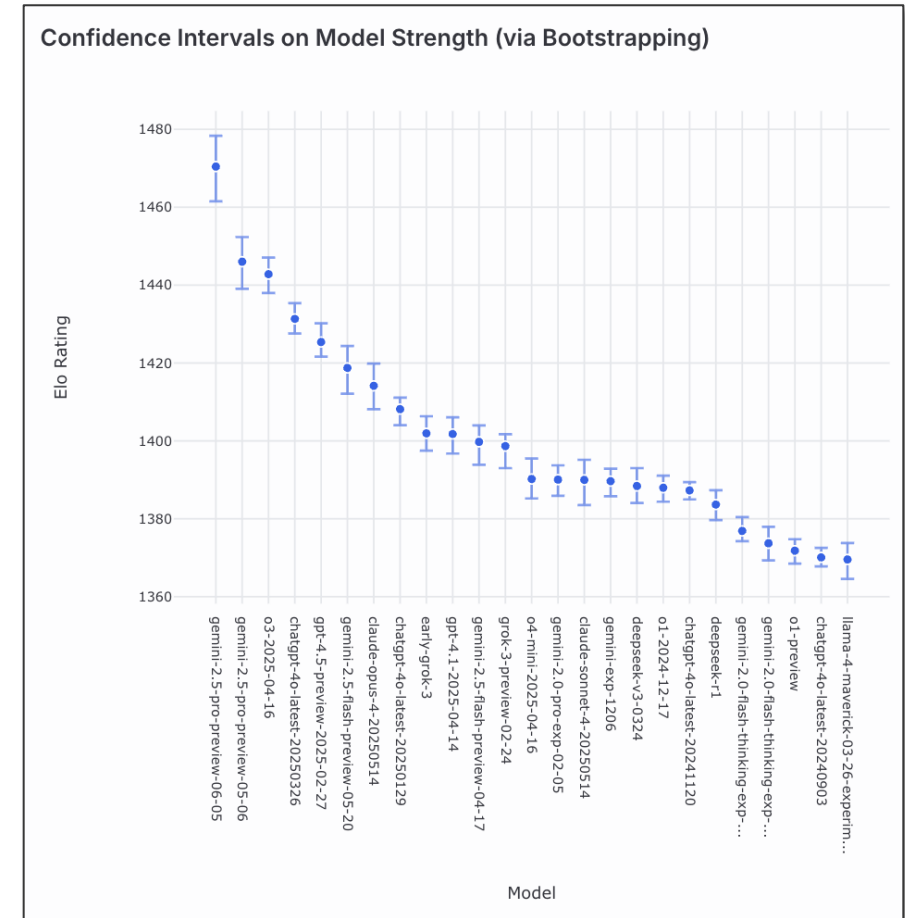
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Using Battles to Generate a Preference Leaderboard

Bradley-Terry Model (next week): estimate the relative strengths of models (**X**) through logistic regression (**Y** = winner)

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^M} \frac{1}{n} \sum_{i=1}^n \text{BCELoss}(\text{sigmoid}(X_i^\top \beta), Y_i)$$

The **Arena Score** of a model is its coefficient in this model (β)



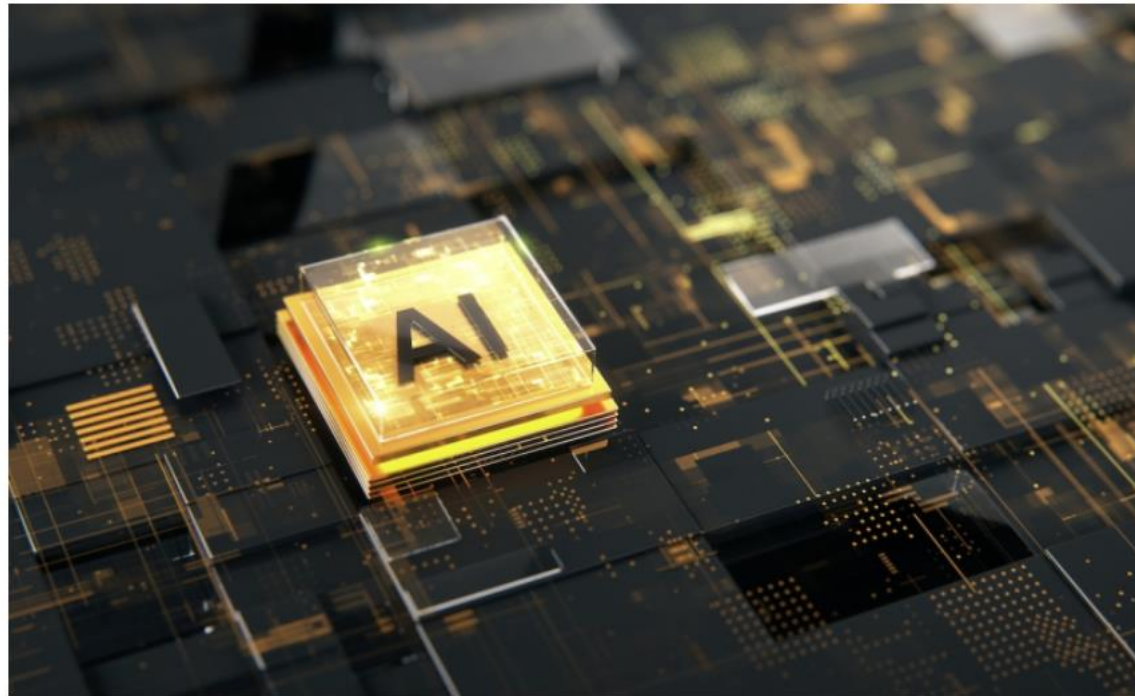
If you think logistic regression is not valuable



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Chatbot Arena Raises \$100 Million for Platform That Compares AI Models

BY PYMNTS | MAY 21, 2025



[Chatbot Arena](#) raised \$100 million in seed funding to support its platform that helps users compare the performance of different artificial intelligence models.

The company also said in a Wednesday (May 21) [post](#) on social platform X that it plans to relaunch the platform, [LMArena](#), with a new look based on feedback from members of its community.



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Structure of a paper

- Abstract
- Intro
- Related works
- Methods and/or Problem Formulation
- Results
- Limitations and Conclusion



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What to answer in every paper

1. What problem is this paper tackling?
2. What do prior works do and how do they fall short?
3. What is the key insight of this paper? What do they do that prior works do not?
4. What are the inputs and outputs of their method/contribution?
5. What are limitations to this method?

1-4 can usually be found in the intro



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What to answer in every paper

1. What problem is this paper tackling? 1st paragraph
2. What do prior works do and how do they fall short? 1st - 2nd
3. What is the key insight of this paper? What do they do that prior works do not? 2nd- 4th
4. What are the inputs and outputs of their method/contribution? 2nd- 4th
5. What are limitations to this method?

1-4 can usually be found in the intro